PHOT 451: Microscale optical system design
LECTURE 01

Michaël Barbier, Fall semester (2026-2027)

COURSE INFORMATION

Instructor

Dr. Michaël Barbier
e-mail: michaelbarbier@iyte.edu.tr
Office: Z-5/B
Hours: Friday 9:00 - 17:00 (via appointment)

Teaching Assistant(s)

Eray Ceyhan
e-mail: erayceyhan@iyte.edu.tr
Office: Z-9/B
Hours: TBA

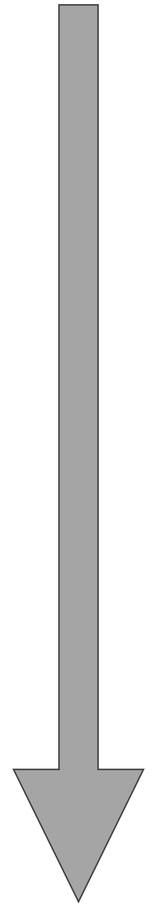
Course Schedule

Monday	13:30 – 15:15 (theory)	Building F, Ali Küçük lab (computer lab first floor)
Monday	15:30 – 17:15 (practice)	Building F, Ali Küçük lab (computer lab first floor)

CONTENTS OF THE COURSE

- Review:
  - Microscale lenses, gratings, binary vs. blazed
  - Math: Complex numbers, Matrices, Fourier transforms
  - Wave propagation: Free space, wave guides, fibers
  - Approximations & Aberrations

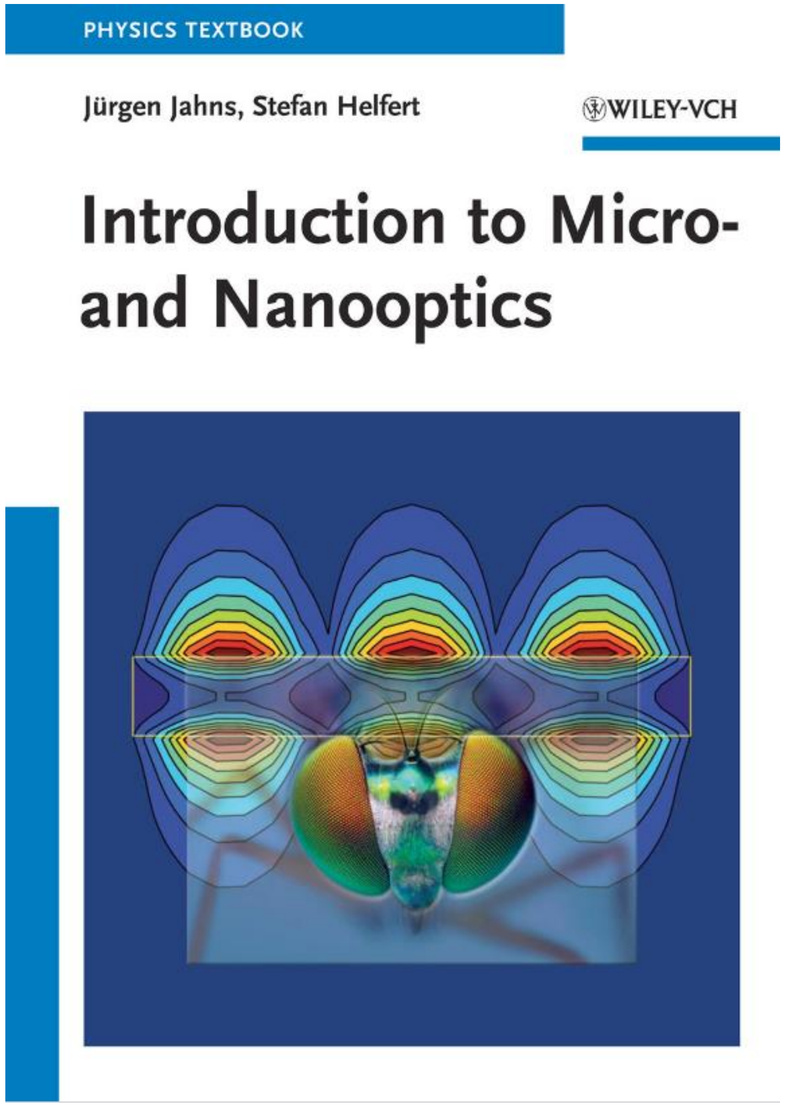
- Optics:
  - Refractive: lenses (Reflective optics)
  - Diffractive optics
- Microlens arrays
- Wave guides, Coupled wave guides
- Photonic crystals



COURSE MATERIALS

Course book

J. Jahns, S. Helfert, Introduction to Micro- and Nanooptics, Wiley (2012)



Supplementary materials

S. Bhattacharya, **Introduction to Ray, Wave, and Beam Optics with Applications**, IOP (2024)

R. Siew, **Modern Classical Optical System Design**, IOP (2024)

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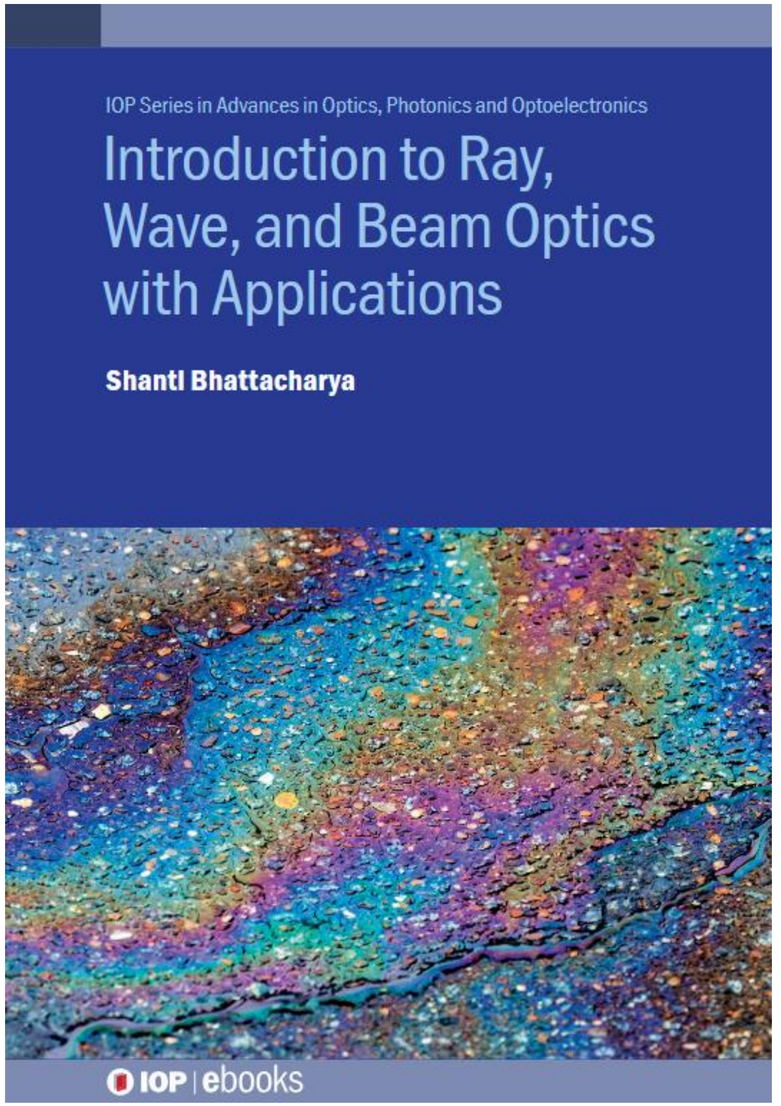
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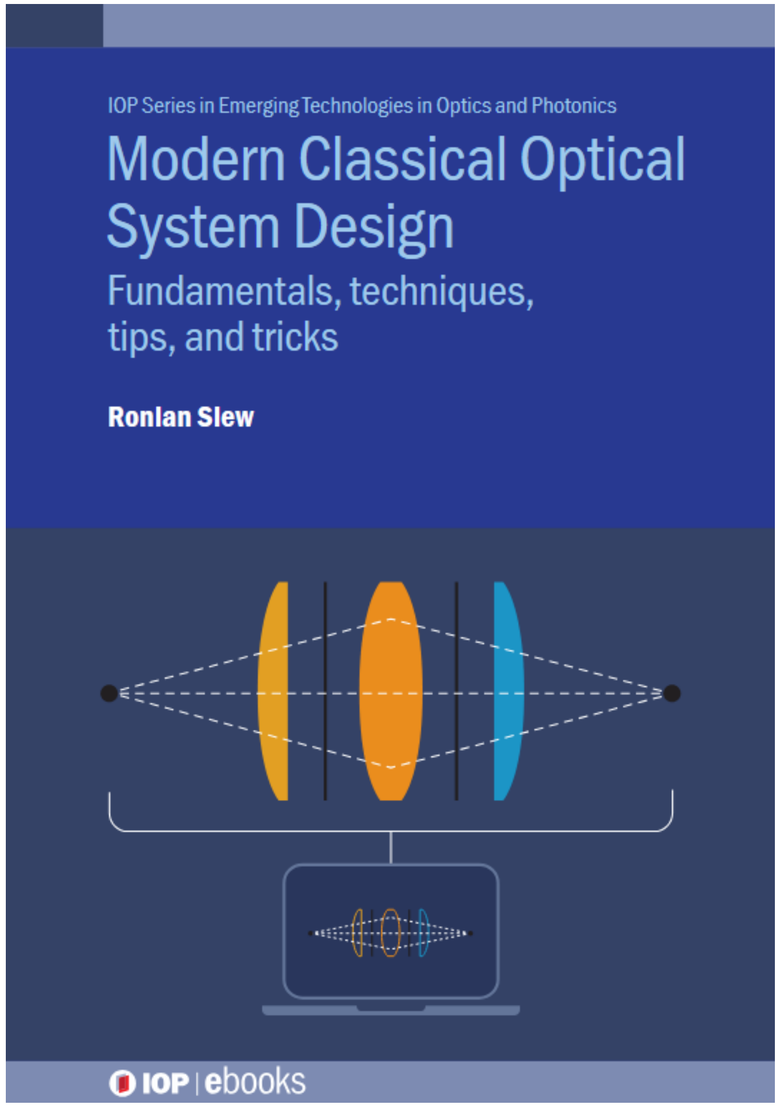
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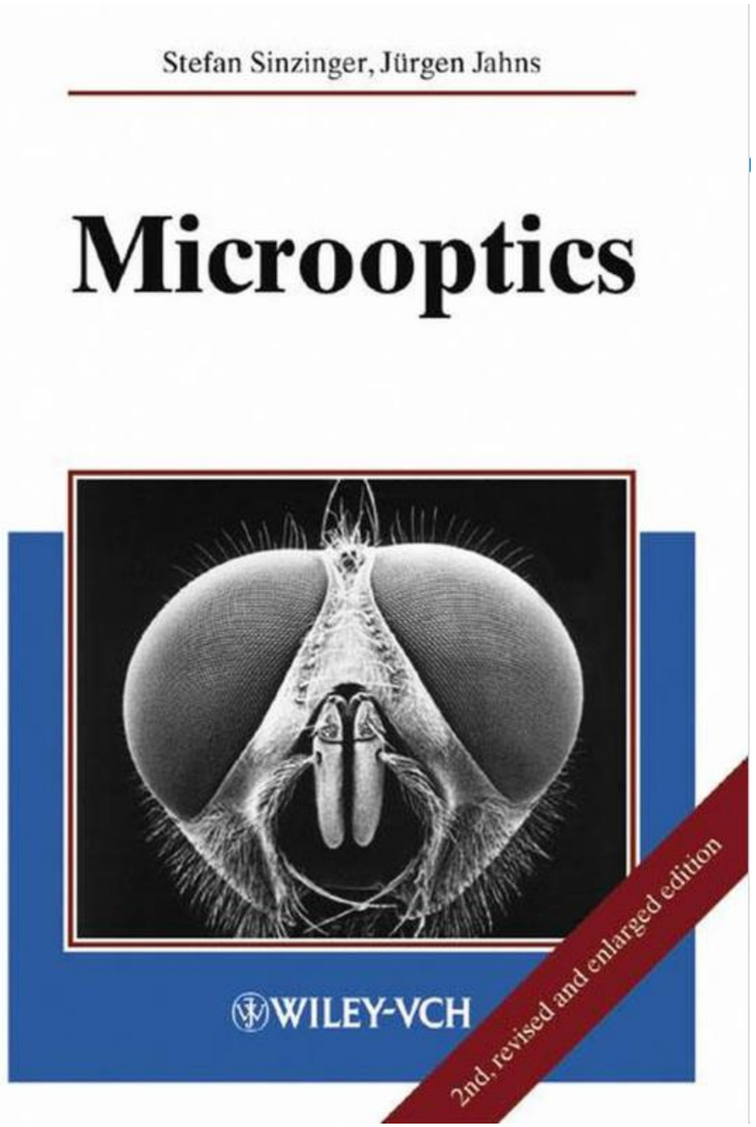
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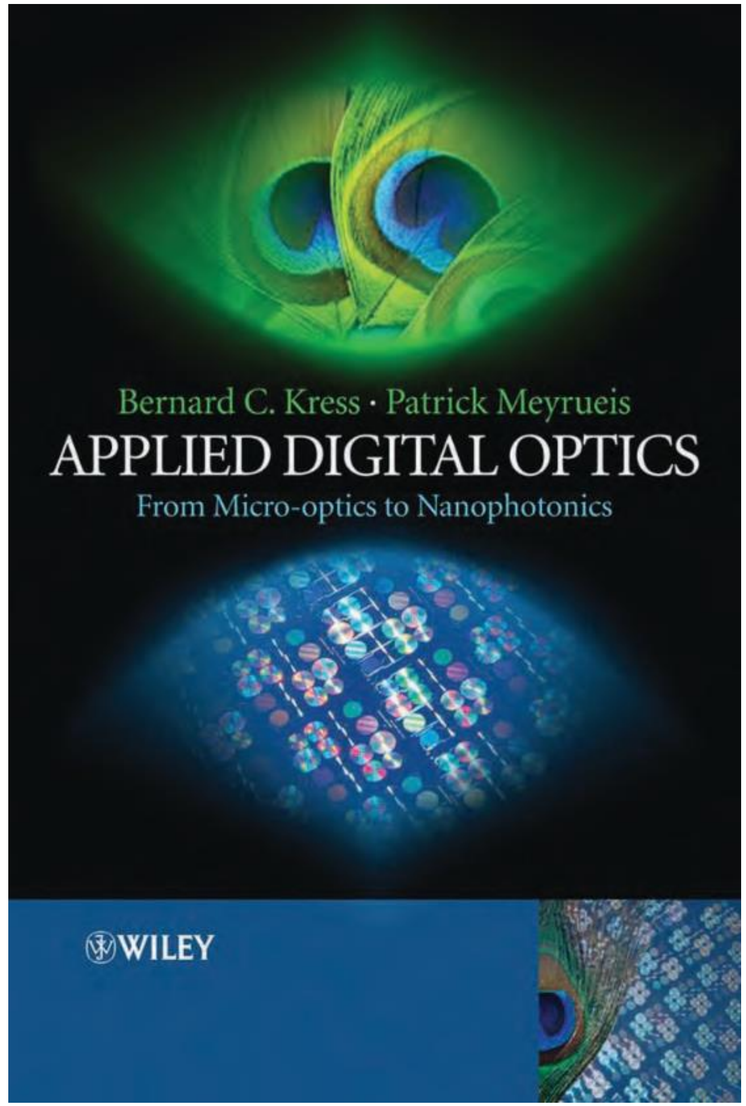
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COURSE MATERIALS

Supplementary material

Youtube channel on design of optical systems using Zemax

Python API for Lumerical

ZOS-API for Zemax (connect via MATLAB, Python, C#, and C++)

OVERVIEW OF THE COURSE

week	Topic
Week 1	Introduction to micro-scale optical components
Week 2	Light propagation in free space
Week 3	Geometric optics and raycasting
Week 4	Diffraction limit & Abberations
Week 5	Beam propagation
Week 6	Refractive optical elements Microlenses
Week 7	Blazed Fresnel lenses
Week 8	Digital lenses
Week 9	Diffractive optical elements
Week 10	**Project presentations**
Week 11	Wave guides and beam propagation
Week 12	Wave mixing
Week 13	photonic crystals, Gratings, periodic structures
Week 14	Whole optical system optimization

COURSE SYLLABUS AND CLASS WORKFLOW

Homeworks

- 20% of your grade

1 project with presentation

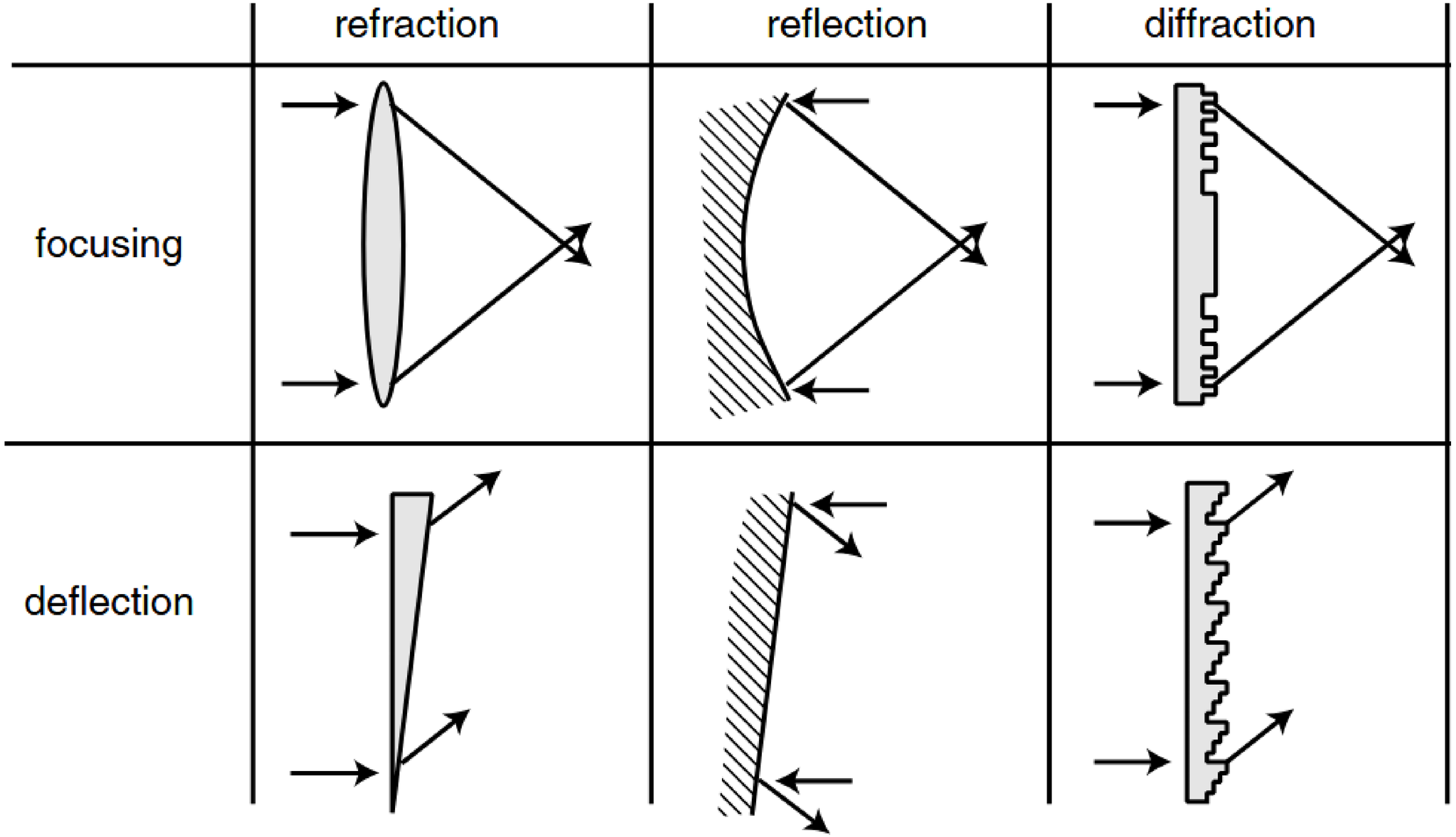
- 30% of your grade

Final Exam

- 50% of your grade
- Topics from the whole year
- Duration: 3 hours

Towards Micro-optical Elements

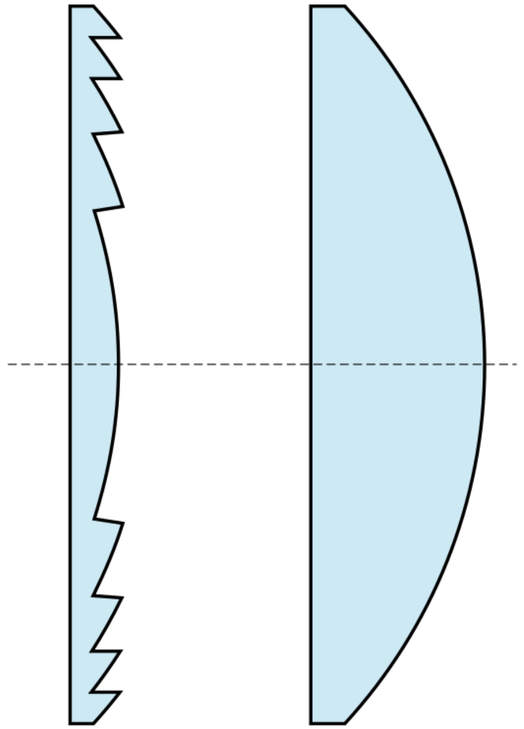
OPTICAL ELEMENTS

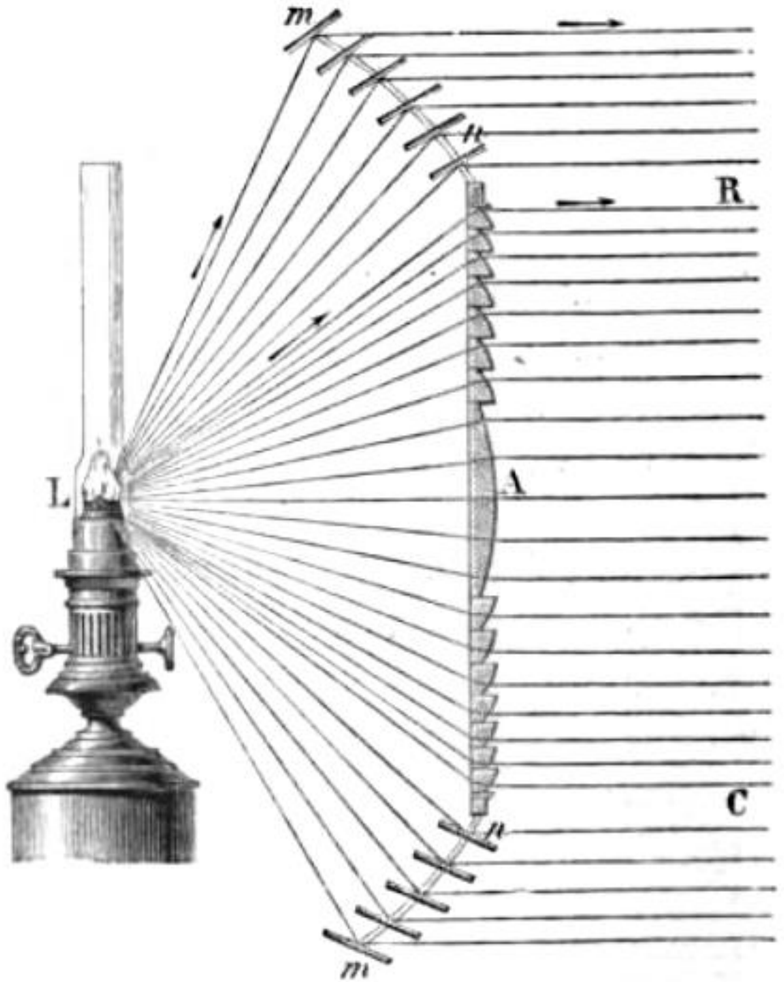


Stefan Sinzinger, Jürgen Jahns, Microoptics, 2nd edition Wiley (2003)

OPTICAL ELEMENTS

- Fresnel Lens



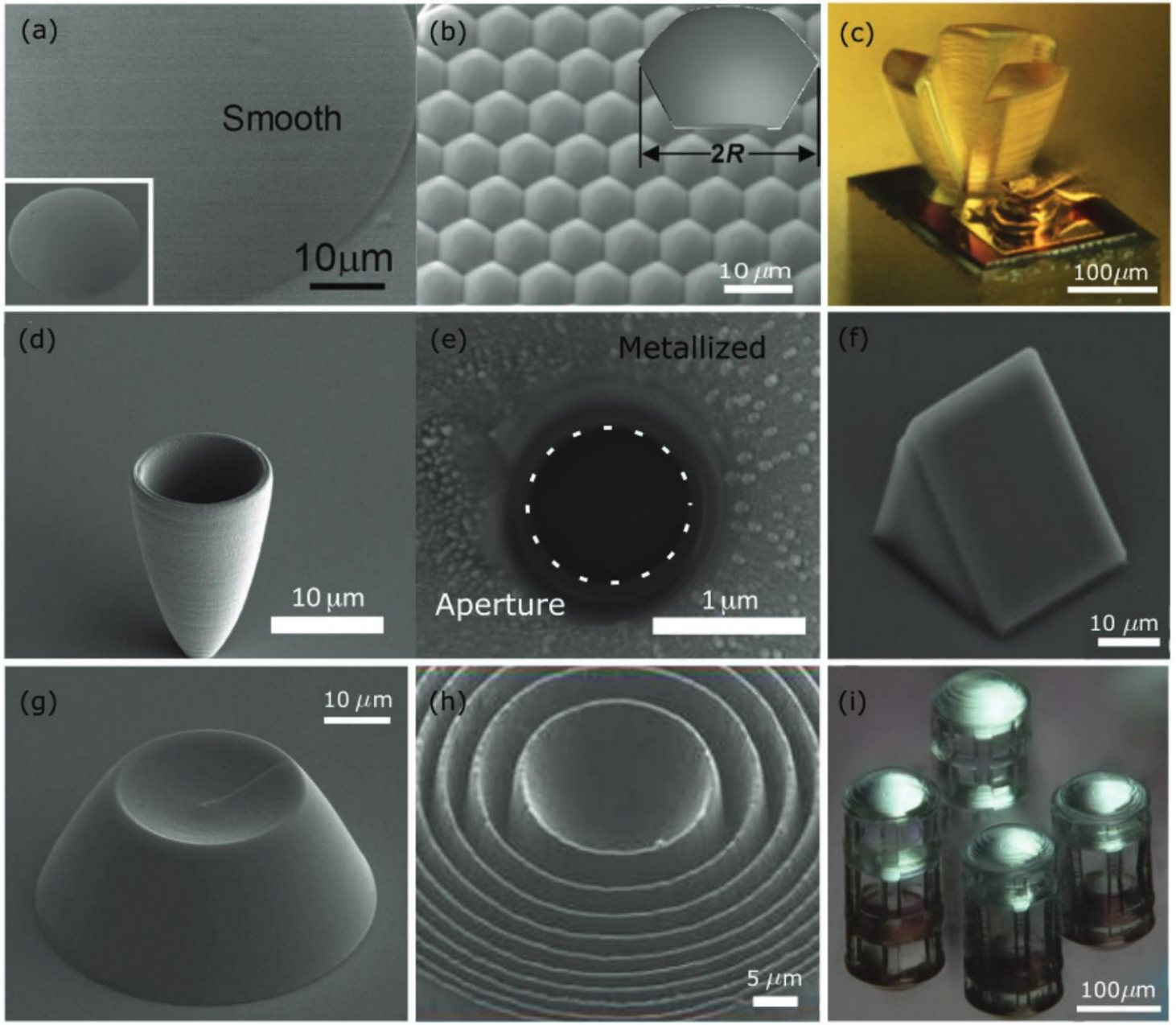




1821 Augustin-Jean Fresnel

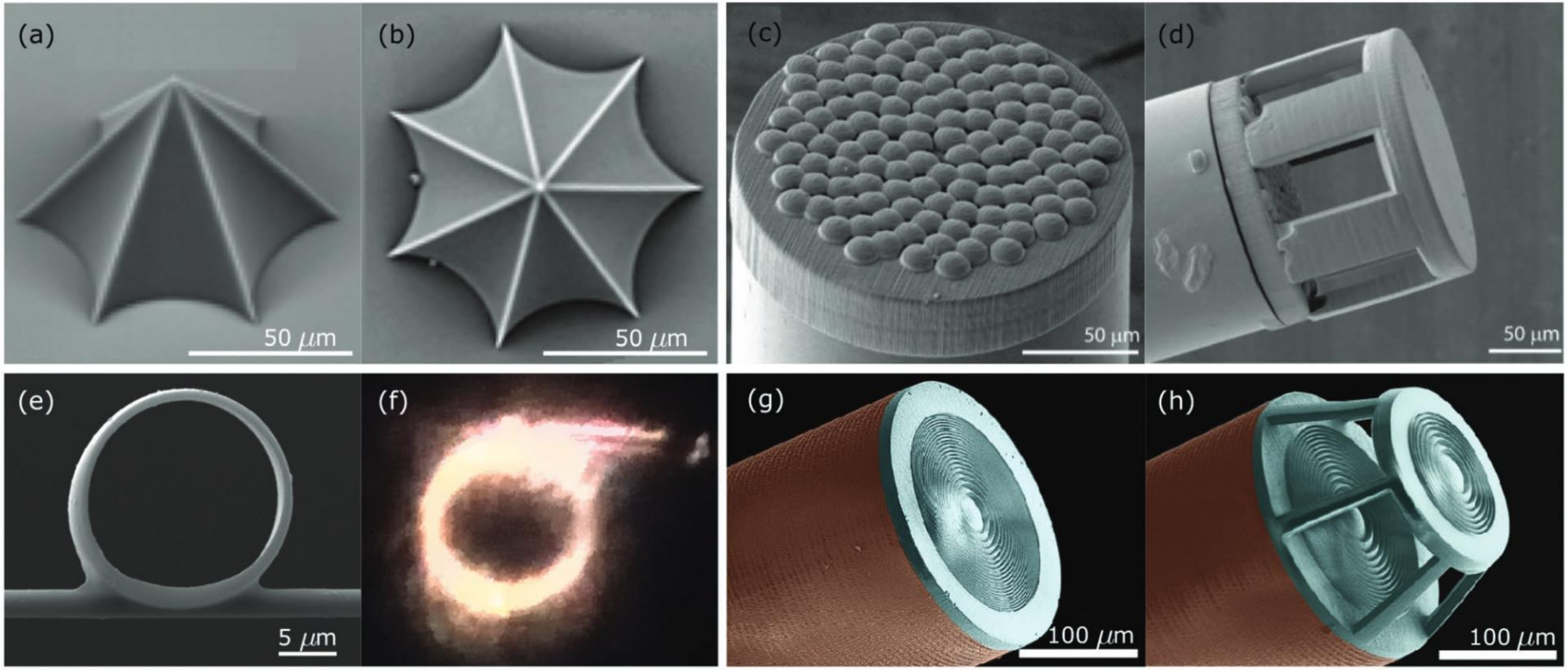
Images adapted From Wikipedia

MICRO-OPTICAL ELEMENTS



Micro-Optics 3D Printed via Multi-Photon Laser Lithography , Diana Gonzalez-Hernandez et al.
Adv. Optical Mater. 2023, 11, 2201701

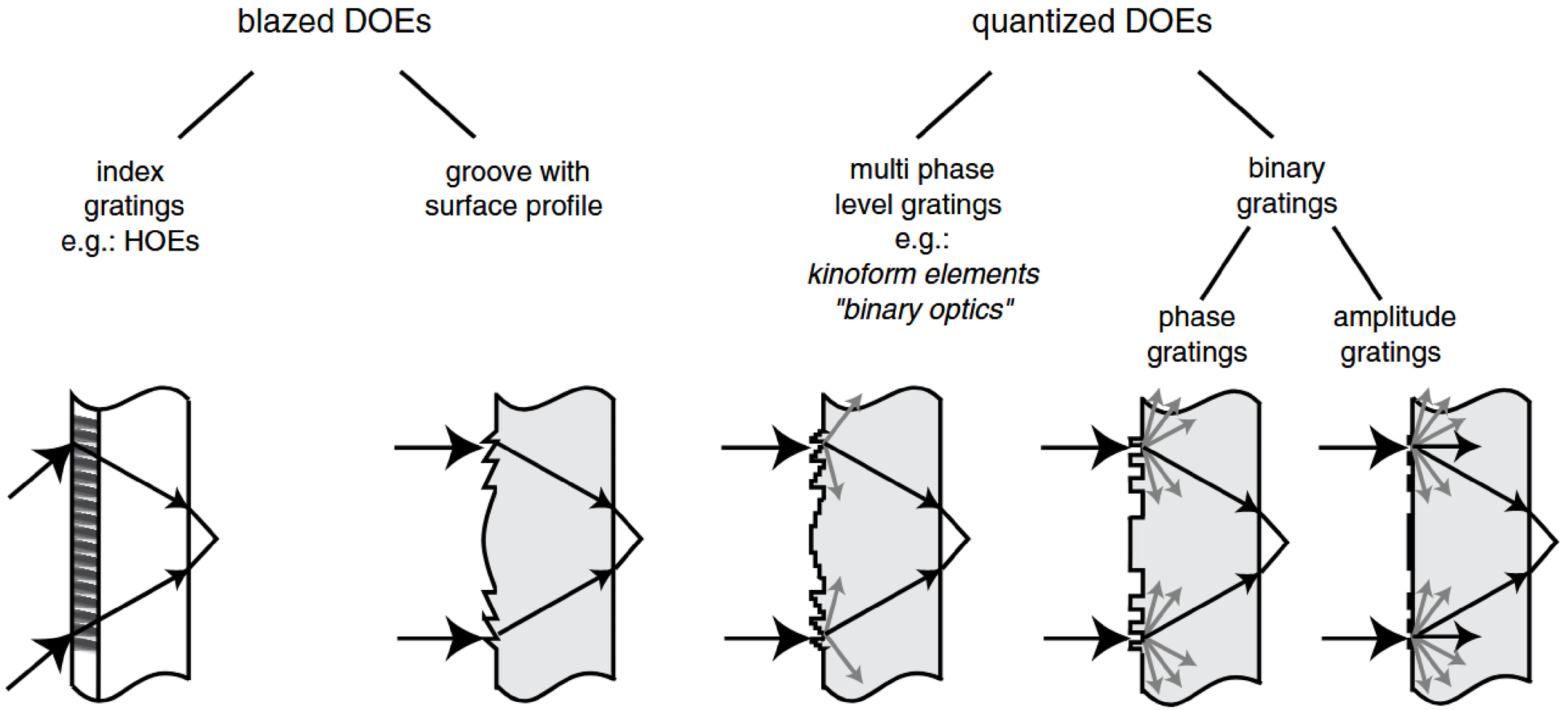
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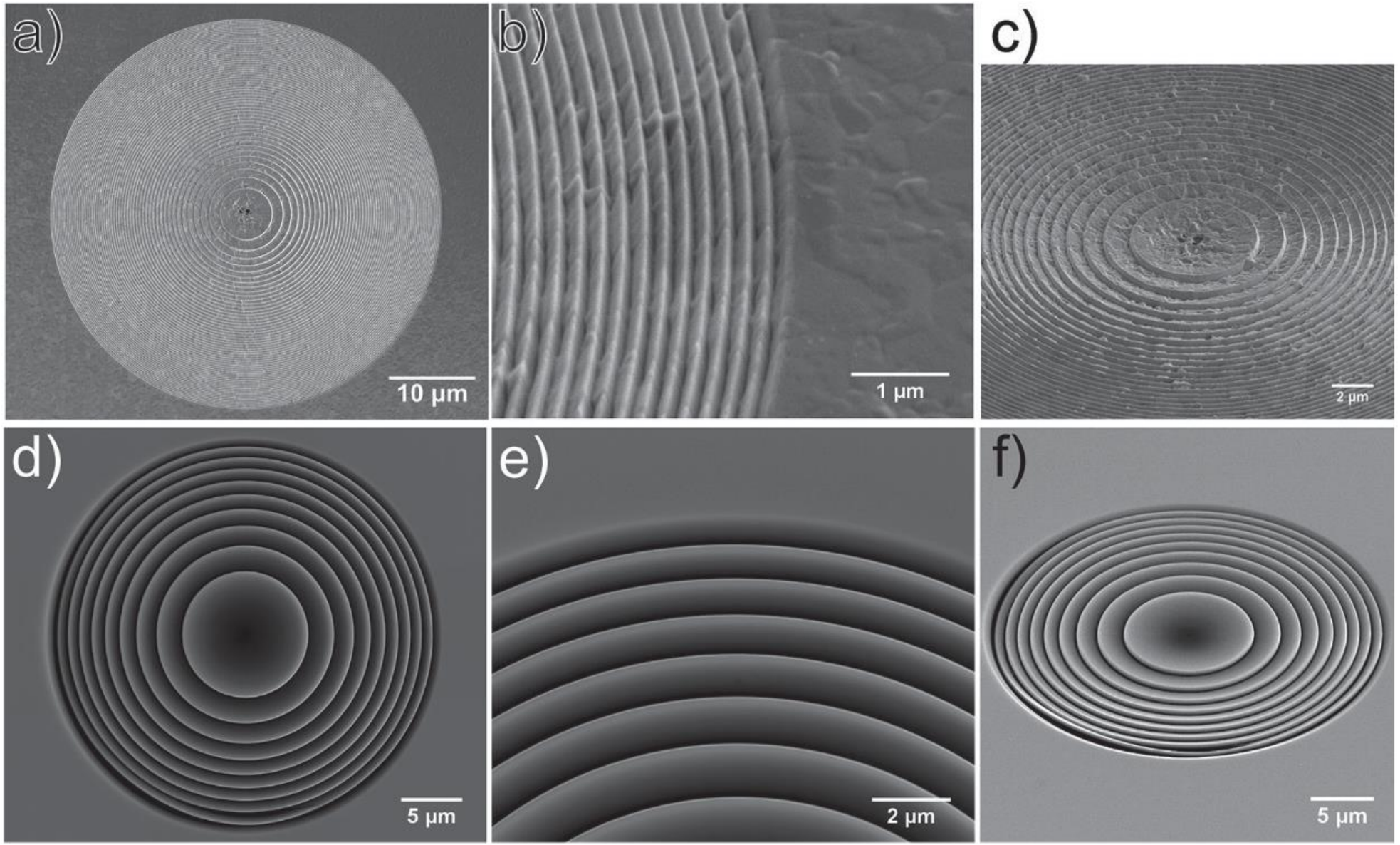
BLAZED, QUANTIZED & BINARY

- Diffractive Optical elements



Stefan Sinzinger, Jürgen Jahns, Microoptics, 2nd edition Wiley (2003)

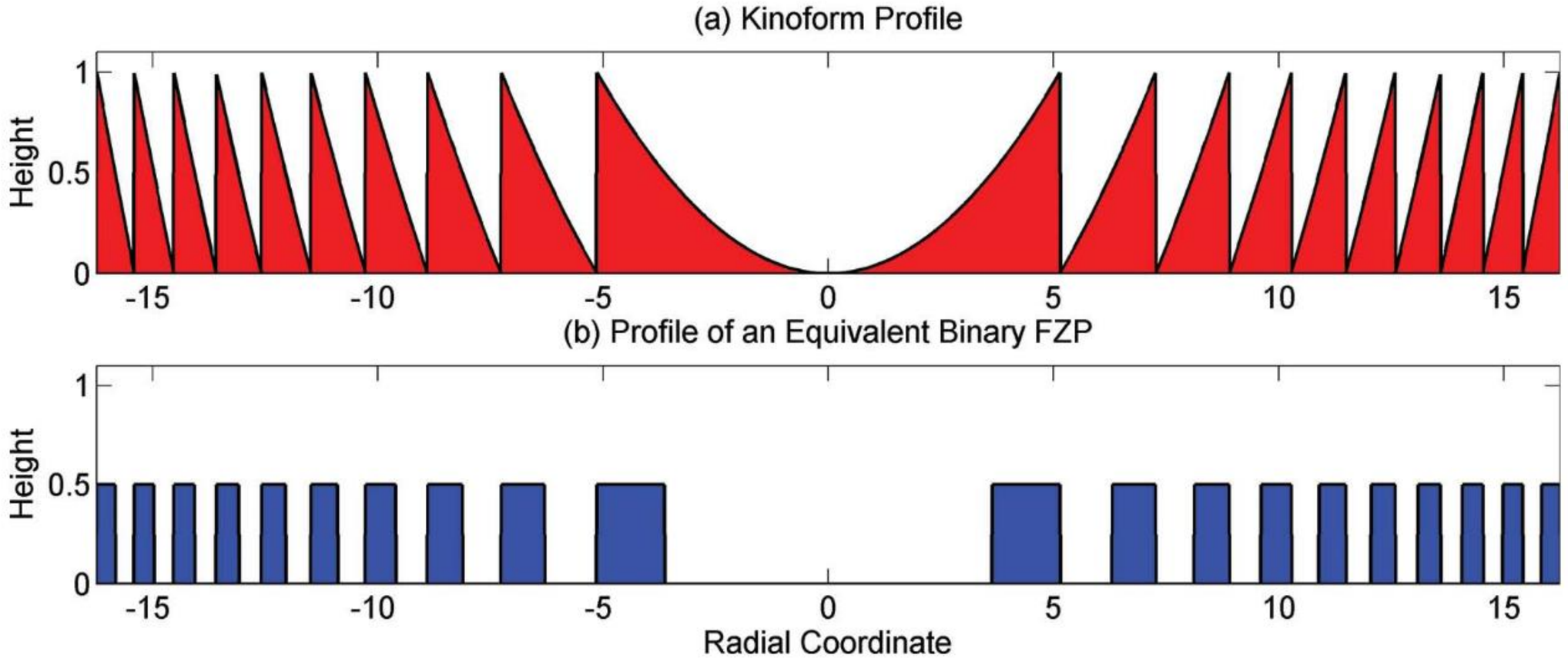
BLAZED VS. BINARY



Kahraman Keskinbora et al., Adv. Optical Mater. 2015, 3, 792–800

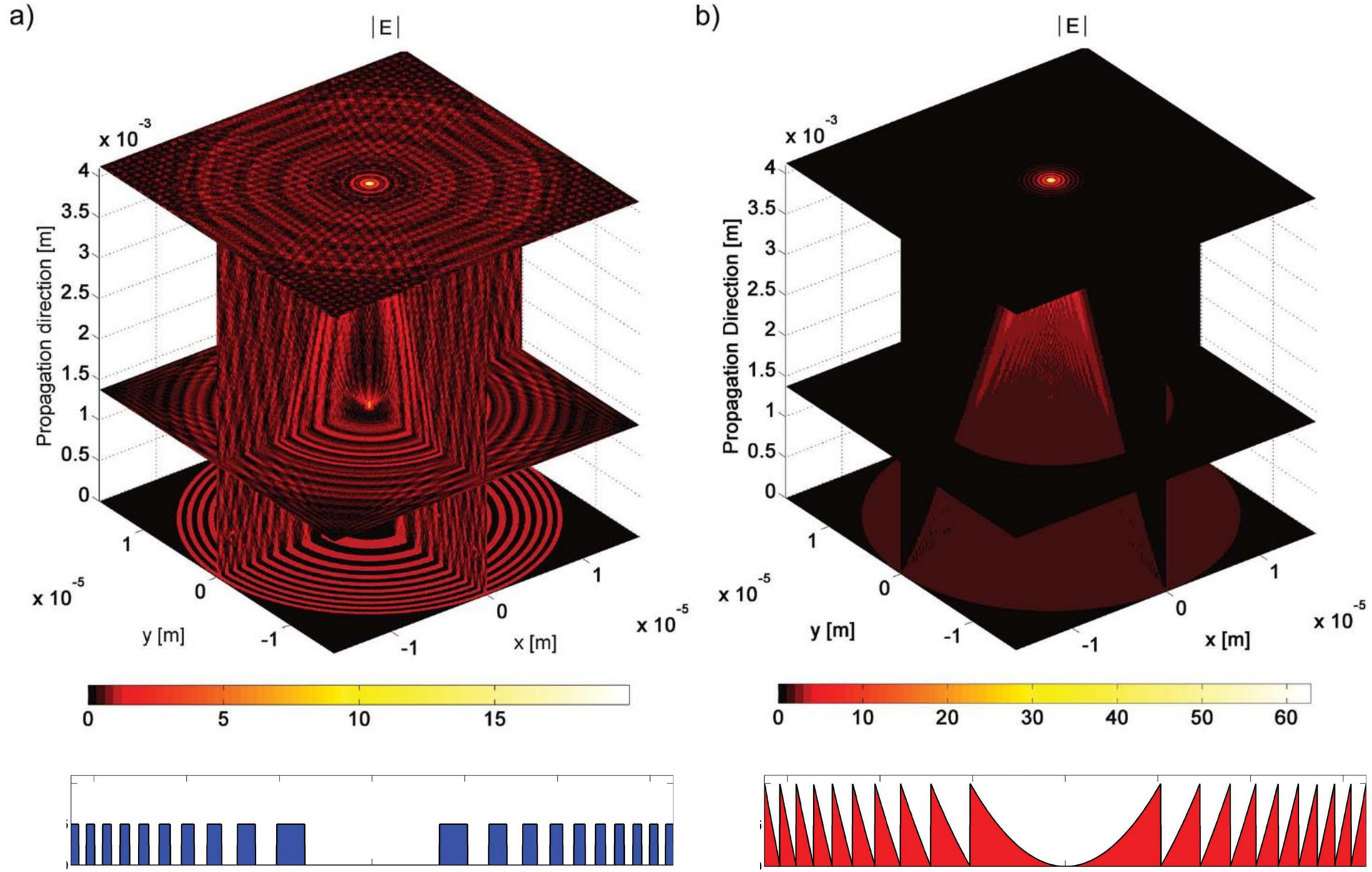
BLAZED VS. BINARY

- Blazed harder to manufacture but better performance



Kahraman Keskinbora et al., Adv. Optical Mater. 2015, 3, 792–800

BLAZED VS. BINARY



Kahraman Keskinbora et al., Adv. Optical Mater. 2015, 3, 792–800

FROM MACRO TO MICRO

- Scaling down optical elements
- Smaller elements → larger artifacts
- Wavelength vs. object size:
  - Geometric ray optics ⟸ Simulate by Raycasting (e.g. Zemax)
  - Scalar wave equation ⟸ Uncoupled electric & magnetic field
  - Paraxial approximation
  - Beam optics ⟸ Beam propagation, ABCD matrices
  - Electromagnetic waves ⟸ Electromagnetic waves, simulate by FDTD (e.g. Lumerical)